

Problem 1 [TODO]

Let $a_1, a_2, \dots, a_n$ be elements of a group G . Define the product of the a_i 's by induction:

$$a_1 a_2 \cdots a_n = (a_1 a_2 \cdots a_{n-1}) a_n.$$

(a) Prove that

$$a_1 a_2 \cdots a_n b_1 b_2 \cdots b_m = (a_1 a_2 \cdots a_n)(b_1 b_2 \cdots b_m).$$

[TODO]

(b) Prove that $a_1 a_2 \cdots a_n$ is equal to the product of the a_i 's with the parentheses inserted arbitrarily. [TODO]

Solution. (a): $a_1 \cdots a_n b_1 = (a_1 \cdots a_n) b_1$ by the definition of the product above. Suppose for the sake of induction,

$$a_1 \cdots a_n b_1 \cdots b_k = (a_1 \cdots a_n)(b_1 \cdots b_k).$$

Thus by the definition of product and the inductive hypothesis, we have that

$$\begin{aligned} a_1 \cdots a_n b_1 \cdots b_{k+1} &= (a_1 \cdots a_n b_1 \cdots b_k) b_{k+1} \\ &= (a_1 \cdots a_n)(b_1 \cdots b_k) b_{k+1} \\ &= (a_1 \cdots a_n)(b_1 \cdots b_{k+1}). \end{aligned}$$

Thus by induction the claim is true.

(b): The statement is trivially true for $n = 1$. Suppose for the sake of induction, the statement is true for all $n \leq k$.

Problem 2 [TODO]

(a) Prove that for every integer $n > 0$, the set of all complex n th roots of unity is a group with respect to complex multiplication. Show that this group is cyclic. [TODO]

(b) Prove that if G is a cyclic group of order n and k divides n , then G contains exactly one subgroup of order k . [TODO]

Solution. (a): Let $\omega = e^{2\pi i/n}$. Let $\mu_n := \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$, is the set of complex n -th roots of unity. $\forall a, b \in \mu_n$, $a = \omega^\alpha$ and $b = \omega^\beta$ for some $\alpha, \beta \in \{0, \dots, n-1\}$. Then

$$\begin{aligned}
ab &= \omega^{\alpha+\beta} \\
&= e^{2\pi i(\alpha+\beta)/n} \\
&= \begin{cases} e^{2\pi i(\alpha+\beta)/n} & \text{if } \alpha + \beta < n \\ e^{2\pi i(\alpha+\beta-n)/n} & \text{if } \alpha + \beta \geq n \end{cases} \in \mu_n.
\end{aligned}$$

$1 * a = a$, by complex multiplication. Furthermore, $a = \omega^{2\pi i \alpha / n}$, then let $b = \omega^{2\pi i (n-\alpha) / n}$. Then $ab = e^{2\pi i} = 1$, and same with ba . Thus μ_n is a group. Furthermore, by construction $\forall a \in \mu_n$, $a = e^{2\pi i \alpha / n} = \omega^\alpha$. Thus ω generates μ_n .

(b): Let G be a cyclic group of order n , thus $\exists x \in G$ such that $x^n = 1$ (but $x^\alpha \neq 1$ for all $0 < \alpha < n$) and $\forall a \in G$, $a = x^\alpha$ for some $\alpha \in \{1, \dots, n-1\}$. Suppose $k \mid n$. Thus $\exists m \in \mathbb{Z}$ where $km = n$. Consider the subgroup $H = \langle x^m \rangle \subseteq G$. By definition H is cyclic, furthermore, $(x^m)^k = x^{mk} = x^n = 1$. Suppose $\alpha \in \{1, \dots, k-1\}$, then $(x^m)^\alpha = x^{m\alpha} \neq 1$ because $0 < m\alpha < n$. Thus $|H| = k$.

Problem 3 [TODO]

- (a) Show that if K and N are two finite subgroups of a group G of relatively prime orders, then $K \cap N = \{e\}$. [TODO]
- (b) Show that if a group G has only a finite number of subgroups, then G is finite. [TODO]

Proof: (a): Suppose $K, N < G$ subgroups where $\gcd(|K|, |N|) = 1$. Note that $K \cap N$ is a subgroup of K and N . We know that $|K \cap N|$ divides $|K|$ and $|N|$, though since the greatest common divisor is 1, $|K \cap N| = 1 \Rightarrow K \cap N = \{e\}$.

(b) □

Problem 4 [TODO]

Prove that a group G is cyclic if and only if there is an element $a \in G$ with $\text{ord}(a) = |G|$.

Proof: $\Rightarrow$: Suppose G is cyclic. By definition, $\exists x \in G$ such that every element of G can be realized as a integer power of x . Let $k = \text{ord}(x)$.

1. $k < \infty$: $\forall a \in \mathbb{Z}$, let $r, p \in \mathbb{Z}$ where $r \in \{0, \dots, k-1\}$ and $pk + r = a$. Then

$$x^a = x^{pk+r} = (x^k)^p x^r = 1^p x^r = x^r.$$

Thus $\{1, x, \dots, x^{k-1}\} = G$ and $|G| = k$.

2. $k = \infty$: Then note that $\forall a, b \in \mathbb{Z}$ where $a \neq b \Rightarrow x^a \neq x^b$ otherwise $x^{a-b} = 1$ and x has finite order. Then $\{x^z : z \in \mathbb{Z}\} \subseteq G$ and $G \subseteq \{x^z : z \in \mathbb{Z}\}$ thus $G = \{x^z : z \in \mathbb{Z}\} = \langle x \rangle$

$\Leftarrow$: Suppose $\exists x \in G$ where $k = \text{ord}(a) = |G|$.

1. $k < \infty$: $S = \{1, x, \dots, x^{k-1}\}$ has k distinct elements of G and $S \subseteq G$. But since $|G| = k$. $S = G$. Thus x generates G and G is cyclic.

2. $k = \infty$: Statement is false in this case. Consider $G = \mathbb{Z}^2$. Both $\mathbb{Z}$ and G are countably infinite, but G has two generators (hence not cyclic).

□

Problem 5 [TODO]

- (a) Show that if $a^2 = e$ for all elements a of a group G , then G is abelian. [TODO]
- (b) Prove that if G is a finite group of even order, then G contains an element a such that $a^2 = e$ and $a \neq e$. [TODO]
- (c) Show that every subgroup of index 2 is normal. [TODO]

Solution. (a): Suppose $\forall a \in G$ we have that $a^2 = e$. Then $\forall a, b \in G$ we have

$$\begin{aligned}(ab)^2 &= abab = 1 \\ ab &= b^{-1}a^{-1} = ba\end{aligned}$$

Thus G is abelian.

(b): Suppose G is a finite group with even order. Suppose $g \in G$ where $\text{ord}(g) > 2 \Rightarrow g^{-1} \neq g$. Let $O_n = \{g \in G : \text{ord}(g) = n\}$ and $P_n = \{g \in G : \text{ord}(g) > 2\}$. Thus $G = O_1 \sqcup O_2 \sqcup P_2$. Note $|O_1| = 1$ and $|P_2|$ is even because if $a \in P_2$ then $a^{-1} \in P_2$. Thus if $|G|$ is even, then $|O_2|$ must be odd, equivalently $|O_2| \geq 1$. Thus $\exists g \in O_2 \subset G$ where $g \neq e$ and $g^2 = e$.

(c): Let $H \leq G$ where $[G : H] = 2$. Let $g \in G - H$, $G/H = \{H, gH\}$ and $G \setminus H = \{H, Hg\}$ since $[G : H] = 2$. $\forall h \in H$, $gh \notin H$, thus $gh \in Hg \Rightarrow \exists h' \in H$ where $gh = h'g \Rightarrow ghg^{-1} = h'$. Thus $gH = Hg \forall k \in G$:

1. $k \in H$: $kHk^{-1} = H$ since H is a subgroup.
2. $k \notin H$: Thus $k \in gH$ and $k = ga$ for $a \in H$. $\forall h \in H$,

$$khk^{-1} = gah(ga)^{-1} = gahag^{-1} = gh'g^{-1} \in H$$

for $h' = aha^{-1}$.

Thus H is normal.

Problem 6 [TODO]

Find all groups (up to isomorphism) of order ≤ 5 . What is the smallest order of a non-cyclic group?

Solution. I will use as fact that any group with prime order is cyclic. Thus the only groups with order 2, 3, and 5 are $\mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$ respectively. The interesting question is studying groups of order 4. Let G be a group of order 4.

1. G cyclic: $G = \mathbb{Z}_4$

2. G is not cyclic: There exists a element $a \in G$ where $\text{ord}(a) = 2$. Then $H = \langle a \rangle$ is a index 2 subgroup of G and hence normal. Suppose $b \notin H$, then $|bH| = 2$ ie $bH = \{b, ba\}$. Note that $\text{ord}(b) \mid 4$ but $\text{ord}(b) \neq 4$ or 1 (because then $b = e \in H$ or G cyclic). Thus b has order 2. Let $\varphi : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow G$ where $(1, 0) \rightarrow a$ and $(0, 1) \rightarrow b$. Suppose $\varphi(k_1, k_2) = e \Rightarrow a^{k_1} b^{k_2} = e \Rightarrow a^{k_1} = b^{-k_2}$. Note that $b \notin H$, thus $k_2 = 0$. Thus $a^{k_1} = e \Rightarrow k_2 = 0$. Thus φ injective. Thus $\mathbb{Z}_2 \times \mathbb{Z}_2 \leq G$. But the orders are the same thus $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Problem 7 [TODO]

Find all subgroups of the symmetric group S_3 . Determine which subgroups are normal.

Solution. $S_3 = \langle x, y : x^2 = y^3 = 1, xyx = y^{-1} \rangle$. The subgroups of S_3 are $e, \langle x \rangle, \langle xy \rangle, \langle xy^2 \rangle, \langle y \rangle$, and S_3 . e, S_3 are both trivially normal. Suppose H is a normal subgroup of S_3 , $|H| = 1, 2, 3, 6$. If $|H| = 1 \Rightarrow H = e$, if $|H| = 6 \Rightarrow H = S_3$ both of which are trivially normal. Since H is normal it is a kernel of a group homomorphism, ie $\varphi : G \rightarrow G/H$ is a group homomorphism. G/H has order $6/|H|$. If $|H| = 3$, then $[G : H] = 2$ and hence H is normal, the order 3 subgroup of S_3 is $\langle y \rangle$. If $|H| = 2$, then $G/H \cong \mathbb{Z}_3$. $S_3 - H$ has 2 more elements of order 2 which must map to something with order 3 which is an impossibility. Thus there is no normal subgroup with order 2 in S_3 . Hence the normal subgroups of S_3 are e, S_3 , and $\langle y \rangle \cong \mathbb{Z}_3$.

Problem 8 [TODO]

Let n be a natural number. Show that the map

$$f : \mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z}, \quad f(a + \mathbb{Z}) = na + \mathbb{Z}$$

is a well-defined homomorphism. Find $\text{Ker}(f)$ and $\text{Im}(f)$.

Solution. **Well Defined:** $f((r + z) + \mathbb{Z}) = n(r + z) + \mathbb{Z} = nr + nz + \mathbb{Z} = nr + \mathbb{Z} = f(r + \mathbb{Z})$ hence f is well defined.

Homomorphism: $f((r + \mathbb{Z}) + (q + \mathbb{Z})) = f((r + q) + \mathbb{Z}) = n(r + q) + \mathbb{Z} = (nr + nq) + \mathbb{Z} = (nr + \mathbb{Z}) + (nq + \mathbb{Z}) = f(r + \mathbb{Z}) + f(q + \mathbb{Z})$

Suppose $f\left(\frac{p}{q} + \mathbb{Z}\right) = \mathbb{Z} \Rightarrow n\left(\frac{p}{q}\right) \in \mathbb{Z} \Rightarrow q \mid n$. Thus

$$\text{ker}(f) := \left\{ \frac{p}{q} + \mathbb{Z} : p, q \in \mathbb{Z}, q \mid n, \gcd(p, q) = 1 \right\}$$

Suppose $\frac{p}{q} + \mathbb{Z} \in \text{im}(f)$, then $\exists r + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ where

$$nr + \mathbb{Z} = \frac{p}{q} + \mathbb{Z} \Rightarrow nr + z = \frac{p}{q} \Rightarrow nr = \frac{p + qz}{q} \Rightarrow r = \frac{p + qz}{qn}$$

Problem 9 [TODO]

- (a) Show that $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. [TODO]
- (b) Prove that $n\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z}$. [TODO]

Problem 10 [TODO]

Let $f : G \rightarrow H$ be a surjective group homomorphism and let H' be a subgroup of H .

- (a) Show that $f^{-1}(H')$ is a subgroup of G . [TODO]
- (b) Prove that the assignment $H' \mapsto f^{-1}(H')$ yields a bijection between the set of all subgroups of H and the set of all subgroups of G that contain $\text{Ker}(f)$. [TODO]
- (c) Prove that a subgroup H' is normal in H if and only if $f^{-1}(H')$ is normal in G . Show that $G/f^{-1}(H') \cong H/H'$. Prove that the assignment $H' \mapsto f^{-1}(H')$ yields a bijection between the set of all normal subgroups of H and the set of all normal subgroups of G that contain $\text{Ker}(f)$. [TODO]